Trenitalia
ETR1000 VZI-6

Drive Control

Revision	Author	Date	Revision Description
00.1	R. Schiavo	30/03/2023	Draft Release Platform.-.Zefiro-Europe/2F_05.01.-.Drive Control-VF.-.Zefiro-Europe/Drive Control-VF.CONCEPT, baseline 3.0 doc 3EST000225-7909 3_A reviewed for VZI-6
00.2	R. Schiavo, G. Iusto	27/04/2023	Draft Release 2F_05.01.Zefiro-Europe.CONCEPT.207" replaced "below" with "above" "2F_05.01.Zefiro-Europe.CONCEPT.184" reported load factor law
00	R. Schiavo	28/04/2023	First Emission
1.1	R. Schiavo	13/11/2023	Modified the following paragraph: Motor traction cooling Added the following paragraphes: Traction Converter cooling, traction converter temperature supervision, traction motor temperature supervision
1.0	R. Schiavo	13/11/2023	Release Emission
2.1	R. Schiavo	20/11/2023	Added propulsion component temperatures to enable Boost mode modified the wrong label .HR15 in .HR55
2.00	R. Schiavo	29/04/2024	Emission – BL1 baseline
3.1	R. Schiavo	28/10/2024	Modified requirements: .184 (Added Ref. Document for weight), .619 (Updated table). .HR40, .HR41,.HR43,. HR44, .HR45,. HR04, .HR55
3.0	R. Schiavo	30/10/2024	Emission
4.1	G.Spina	25/06/2025	Update for MU requirements
4.2	P. Buda	18/07/2025	Modified req: .HR33, .HR34, .HR35, .HR37, .HR38, .HR39. Added req: .HR56
4.3	P. Buda	22/07/2025	Modified req: .HR33
4.0	P. Buda	25/07/2025	Emission BL4 Baseline

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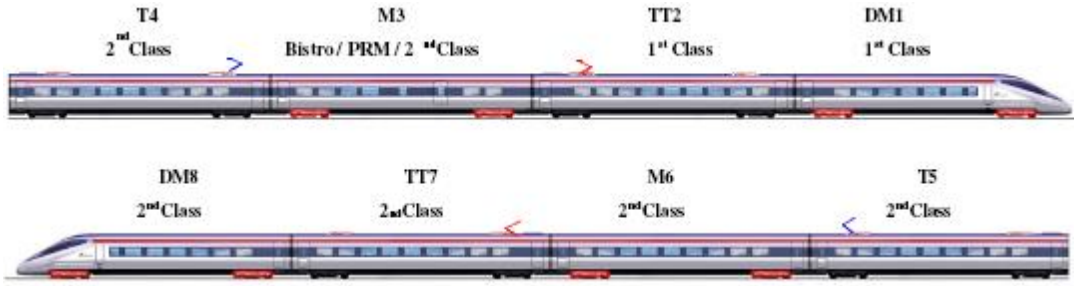
Bus Interface **46**

Introduction

This document is a function specification for the control of the *propulsion* system for V300ZEFIRO for customer Trenitalia.

Train Configuration

Each V300ZEFIRO train consists of eight cars in a fixed formation including a driver's cab at each end.



- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

Definitions and abbreviations

ACM	Auxiliary Converter Module
BCU	Brake Control Unit
CCU	Central Computer Unit
CCU-O	Central Computer Unit – Operation (used for vehicle control)
DC	Direct Current
DCU	Drive Control Unit
DCU/L	Drive control unit line converter
DCU/M	Drive control unit motor converter
DM	End driving motor car with driver's cab
ED	electro dynamic (ED) brake off
EVENT	Diagnostics that will be stored in the Event database within TCMS SW.
FCM	MF-DC Converter Module
HW HMI	Hardware Human Machine Interface, e.g. buttons, switches and indication lamps
ICD	Interface Control Document
ID	Identifier
IDU	Intelligent Display Unit
I/O	Inputs / Outputs
LCM	Line Converter Module
M	Intermediate driving motor car
MCM	Motor Converter Module
MVB	Multifunction Vehicle Bus
N/A	Non Applicable
PCU	Propulsion Control Unit
SFRS	System Functions Requirement Specification
SW	Software
TBD	To Be Defined
TCMS	Train Control and Management System
TDS	Train Diagnostic System
TRD	Technical Requirement Document
TRS	Technical Requirement Specification

T	Intermediate trailer car
TT	Intermediate trailer car with transformer
WSP	Wheel side protection
WTB	Wide Train Bus

System Overview

System Configuration

In each motor car (DM and M cars) there is one complete converter unit.

DM and M cars are equipped with two LCM, one ACM and 4 MCM and two FCM. They are controlled by the PCU Unit which also communicates with TCMS via MVB.

Each LCM controls two traction motors. The communication between TCMS and the propulsion system is achieved through Propulsion Control Unit (PCU). There is one PCU per LCM.

An overview of the electrical power system:

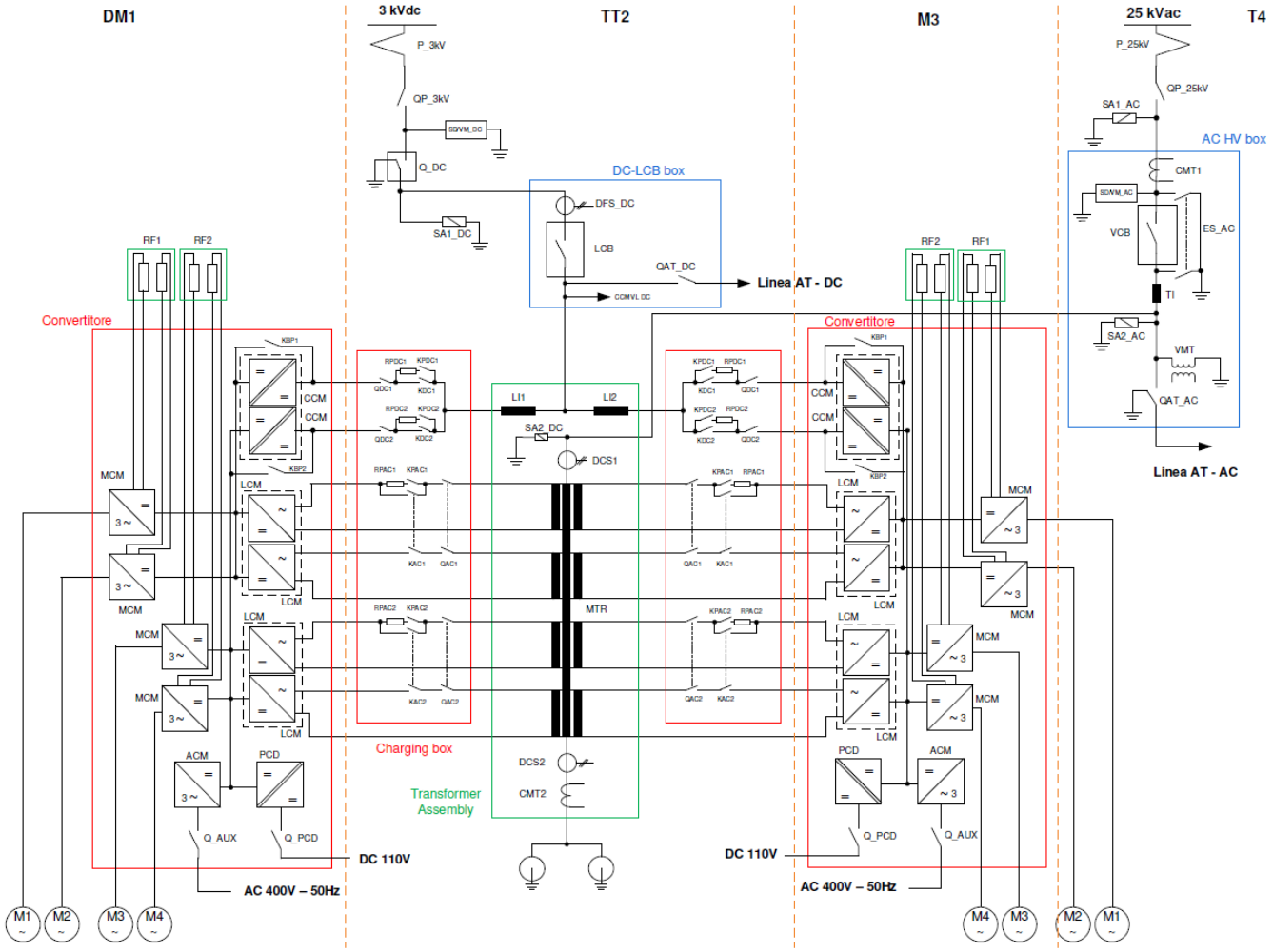


Figure 1 High Voltage Architecture (half Train)

Riferimento schema	Descrizione	Posizione
P_25kV	AC Pantograph(1600mm)	T4, T5 roof
QP_25kV	AC Pantograph Disconnector	T4, T5 roof
SA1_AC	AC Surge Arrester	T4, T5 roof

CMT1,CMT2	AC Line Current Transformer (to energy meter and HVSP)	T4, T5 roof – AC HV box
SD/VM_AC	System Detection to detect and measure the AC line Voltage (to Energy meter, HVSP and Propulsion)	T4, T5 roof – AC HV box
VCB	AC Line Circuit Breaker	T4, T5 roof – AC HV box
ES_AC	AC Isolating/Grounding Switch	T4, T5 roof – AC HV box
TI	Transient Inductor	T4, T5 roof – AC HV box
VMT	AC Line Voltage Transformer (to propulsion)	T4, T5 roof – AC HV box
QAT_AC	AC Train Line Disconnector	T4, T5 roof – AC HV box
P_3kV	DC Pantograph (1450mm)	TT2, TT7 roof
QP_3kV	DC Pantograph Disconnector	TT2, TT7 roof
SD/VM_DC	System Detection (to Propulsion and HVSP)	TT2, TT7 roof
Q_DC	DC System Isolation Switch	TT2, TT7 roof
SA1_DC, SA2_DC	DC Surge Arrester	TT2, TT7 roof
DC CM&VM	DC line current and voltage measurement (to energy meter)	TT2, TT7 roof
DFS_DC	Rogowski current Transducer (to HVSP)	TT2,TT7 Under Frame - DC LCB box
LCB	DC Line Circuit Breaker	TT2,TT7 Under Frame - DC LCB box
QAT_DC	DC Train Line Disconnector	TT2,TT7 Under Frame - DC LCB box
DCS1, DCS2	Transformer primary current transducers (to HVSP – differential current protection)	TT2,TT7 Under Frame – Transformer Box
MTR	25kV Main Tranformer	TT2,TT7 Under Frame – Transformer Box
LI1, LI2	Line Inductor	TT2,TT7 Under Frame – Transformer Box
KDC1, KDC2	DC converter Contactor	TT2,TT7 Under Frame – Charging box
KPDC1, KPDC2	DC Precharge Contactors	TT2,TT7 Under Frame – Charging box
RPDC1, RPDC2	DC Precharge Resistance	TT2,TT7 Under Frame – Charging box
QDC1, QDC2	DC Converter Disconnector	TT2,TT7 Under Frame – Charging box
KAC1, KAC2	AC bipolar converter contactors	TT2,TT7 Under Frame – Charging box
KPAC1, KPAC2	AC Precharge Contactors	TT2,TT7 Under Frame – Charging box
RPAC1, RPAC2	AC Precharge Resistance	TT2,TT7 Under Frame – Charging box
QAC1, QAC2	AC Converter Bipolar Disconnector	TT2,TT7 Under Frame – Charging box
CCM	Chopper Converter Module	DM1,M3,M6,DM8 Under Frame – Converter
LCM	Line Converter Module	DM1,M3,M6,DM8 Under Frame – Converter
MCM	Motor Converter Module	DM1,M3,M6,DM8 Under Frame – Converter
ACM	Auxiliary Converter Module	DM1,M3,M6,DM8 Under Frame – Converter
PCD	Precharge Assembly from 110V	DM1,M3,M6,DM8 Under Frame – Converter
KBP1, KBP2	CCM bypass contactor	DM1,M3,M6,DM8 Under Frame – Converter
QAUX	Auxiliary disconnector	DM1,M3,M6,DM8 Under Frame – Converter
QPCD	Precharge Assembly disconnector	DM1,M3,M6,DM8 Under Frame – Converter
M1, M2, M3, M4	Traction motor	DM1,M3,M6,DM8 -Motor bogie
RF1, RF2	Braking Rheostat	DM1, M3, M6, DM8 - roof

Functional requirements

The traction system configuration is modular consisting of two independent high voltage systems (pantograph, main transformer) and eight independent traction converter systems. The availability of at least 50% of traction power of a train is guaranteed in any single failure case.
To assure 50% traction also in case of fire, separation of the traction systems of the two semi-trains is made possible by isolation contactors.

Driving direction vs. Motor Rotation

Function Design

The driving direction pushbuttons at driver's desk are specified in the SFRS for Conventional Train Control.

In each PCU, the driving direction shall be checked in order to avoid a mismatch with the rotating direction of a motor.

Detected mismatch between driving direction and motor rotation will be indicated by the PCU only as a "mechanical fault" and the respective MCM will be shut off.

Normal condition

MCM shall control that the traction motor rotating direction of each motor is corresponding to the chosen driving direction.

Failure cases

Description of failure cases

A) The wrong direction of the motor is detected.
B) No communication between TCMS and PCU. Not possible to indicate failures to the driver. It is not likely that the motor is rotating in the wrong direction since traction will be disabled at the motors where the communication has been lost between TCMS and PCU. This implies a possible reduction of max 12,5% traction power.
C) The motor is not shut-off when rotating in the wrong direction, i.e. that the protective blocking is not working on the motor. Risk of wheel and motor damage depending on the adhesion.

Failure detection

Description of failure detection

A) The faulty direction is detected by the mechanical fault detection function in the MCM.
B) The communication between CCU and the PCU is supervised by the CCU which will trigger an event stored in the diagnostic database defined in the TCMS Operational SFRS.
C) None

Failure handling

Description of failure handling

A) The MCM is blocked. To be maintained at next scheduled workshop visit.
B) Fault will be logged in the diagnostic database and checked at maintenance. Unscheduled workshop visit at the earliest after two days.
C) Maintenance: Major failure.

Requirements

ID	2F_05.01.Zefiro-Europe.TRS.184
Description	The torque direction exerted by the traction motor (both, in traction or dynamic braking) shall be monitored and compared to the driving direction requested by the driver.
Safety level	SIL 1

ID	2F_05.01.Zefiro-Europe.TRS.186
Description	Driving: In case of discrepancy (traction motor vs. driver request direction), the torque shall be canceled.
Safety level	SIL 1

The rotation direction of the motor shall be supervised and compared to the driving direction acquired from trainlines. In case of discrepancy the motor torque shall be stopped.

Nr	Description	Type	Derived to	User Interface	SIL
2F_05.01.Zefiro-Europe.CONCEPT.207	In the starting phase, in case of mismatch between the rotation of the motor and the demanded driving direction, the motor shall be shut off after short time and above a speed threshold (3km/h) making a hill start possible.	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.615	If a mismatch is detected between the rotation of the motor and the demanded driving direction when the speed is above a threshold (5km/h), an event shall be generated indicating "mechanical fault" and the torque shall be immediately canceled	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)	EVENT	N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR01	If a mismatch is detected between the rotation of the motor and the demanded driving direction in the starting phase when the speed is below a threshold(3km/h), a back status is sent to TCS.	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A

Train Weight Correction Factor for traction

Function Design

The performance of the train shall be independent of its load. To achieve this performance, the traction demand shall take in account the load weigh (passengers, baggage etc.) of the train.

The train mass is measured. The train weight correction factor is directly proportional to the local car weight. The tractive reference is scaled according to the train weight correction factor.

Normal condition

In normal condition, the train weight is measured in each car by the secondary suspension measured by the Braking system in each car. The car weight will be forwarded to TCMS which will calculate the total train weight.
The train weight correction factor will be calculated by TCMS and sent to each PCU which will compensate traction demand accordingly.

Failure cases

Description of failure cases

1. Train weight is invalid.
2. Communication fault between TCMS and PCU. The train weight correction factor will not be sent to the PCU control but that is secondary since the communication fault disables the traction motors where the communication has been lost between TCMS and PCU.
3. Calculated Train weight is less than the physical train weight. This fault will imply that there is an undetected fault in the secondary suspension device.
4. Calculated Train weight is greater than the physical train weight. This will imply that there is an undetected fault in the secondary suspension device.

Failure detection

Description of failure detection

1. The secondary suspension is supervised by the Braking system.
2. The communication between CCU and the PCU is supervised by the CCU which will trigger an event stored in the diagnostic database.
3. Detected by driver due possible problems reaching maximum train speed.
4. No detection available that can be considered as valid.

Failure handling

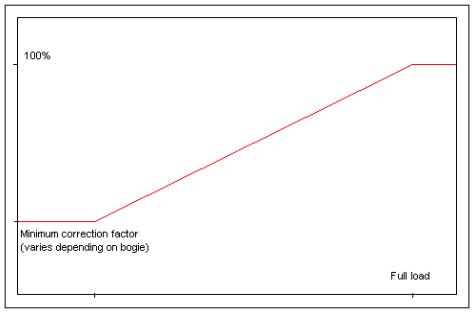
Description of failure handling

1. The train weight correction factor will be set to 100% to allow full performance. An event informing invalid train weight will be set in the diagnostic database.
2. Fault will be logged in the diagnostic database and be checked at maintenance. Unscheduled workshop visit at the earliest after two days.
3. Checked during maintenance based on fault reports by the driver.
4. None identified. The train weight correction factor will be set to 100% to allow full performance.

Requirements

ID	2F_05.01.Zefiro-Europe.TRS.120
Description	The applied tractive force shall be compensated with the current weight of the train.
Safety level	SIL 0

The train mass is measured by the secondary suspension, read by the BCU and forwarded to TCMS software and PCU for load compensation of the traction demand.

Nr	Description	Type	Derived to	User Interface	SIL
2F_05.01.Zefiro-Europe.CONCEPT.70	The car weight as measured by the secondary suspension device shall be forwarded to the TCMS on IP.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.01.Zefiro-Europe.CONCEPT.633	TCMS shall be notified if the measured weight is invalid.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.01.Zefiro-Europe.CONCEPT.184	The train weight correction factor shall be calculated and sent to each PCU in the train. The value shall be between 90-100% where 90% equals service weight and 100% equals normal weight (TSI definition). If the weight is lower than service weight the correction factor shall be 90% and if it is over normal weight the correction factor shall be 100%. The service weight and the Normal weight shall be calculated as reported in the FA020023100 document 2F_05.02.Zefiro-Europe.CONCEPT.1751 requirement, taking into account the table weight (table 1) reported in the same document.  	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.01.Zefiro-Europe.CONCEPT.618	If the car weight cannot be calculated for a car the train weight correction factor shall be set to 100% for that car, i.e. full load shall be considered.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.01.Zefiro-Europe.CONCEPT.636	If the train weight is considered invalid, an event shall be sent.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_05.01.Zefiro-Europe.CONCEPT.185	The traction demand received from the TCMS shall be considered not load compensated. The compensation shall be calculated internally using the train weight correction factor received from TCMS.	Derived Requirement	4S_05.-.Propulsion (Prpulsion Control)		N/A

Car ID and Vehicle Type

Function Design

All traction packages in each motor car work independently from each other.

In order to coordinate all these local propulsion packages, the car ID shall be distributed in all motor cars through the train from TCMS.

Normal condition

In normal conditions, the car ID is distributed from TCMS to all local PCUs which will check the consistency of the received information.

The car ID will be used by the diagnostic function in the PCU.

Failure cases

Description of failure cases

A) The car ID sent to PCU is out of range or mismatch between Car ID and the address acquired from digital input and IP address. Traction will be disabled in each Propulsion module that have invalid car ID.
B) Communication fault between TCMS and PCU. No Car ID will be sent to the PCU.

Failure detection

Description of failure detection

A) If the Car ID is not between 1 and 8 or the Car ID can not be mapped to the PCU address acquired from digital input and IP address.
B) The communication between CCU and the PCU is supervised by the CCU which will trigger an event stored in the diagnostic database.

Failure handling

Description of failure handling

A) Fault will be logged in the diagnostic database and checked at maintenance. Unscheduled workshop visit at the earliest after two days if only one propulsion module is affected. Otherwise: Major failure.
B) Fault will be logged in the diagnostic database and checked at maintenance. Unscheduled workshop visit at the earliest after two days if only one propulsion module is affected. Otherwise: Major failure.

Requirements

ID	2F_05.01.Zefiro-Europe.TRS.104
Description	TCMS shall send a car ID to the Propulsion Control (PCU)
Safety level	SIL 0

The traction packages work independently from each other and need to be informed about the car ID. Each PCU acquires the info about car ID from TCMS, from digital inputs and IP address knows also its position in the car.

<table>
<tr><th>Nr</th><th>Description</th><th>Type</th><th>Derived to</th><th>User Interface</th><th>SIL</th></tr>
<tr><td>2F_05.01.Zefiro-Europe.CONCEPT.105</td><td>TCMS shall send a car ID to the PCU according to the table below:
<table><tr><th>Car identity</th><th>Car description</th></tr><tr><td>1</td><td>DM1</td></tr><tr><td>3</td><td>M3</td></tr><tr><td>6</td><td>M6</td></tr><tr><td>8</td><td>DM8</td></tr></table></td><td>Derived Requirement</td><td>4S_09.03.05.-.TCMS Software</td><td></td><td>SIL 0</td></tr>
<tr><td>2F_05.01.Zefiro-Europe.CONCEPT.619</td><td>If the car ID sent from TCMS is <= 0 or >= 9 or the IP address and /or the address digital input of the PCU does not correspond to the valid Car ID according to below table then an event shall be set.
<table><tr><th>Car identity</th><th>PCU Number</th><th>IP addr</th><th>Digital input coding</th></tr><tr><td>1</td><td>PCU1</td><td>10.0.1.130</td><td>0001</td></tr><tr><td>1</td><td>PCU2</td><td>10.0.1.131</td><td>0010</td></tr><tr><td>3</td><td>PCU3</td><td>10.0.1.132</td><td>0011</td></tr><tr><td>3</td><td>PCU4</td><td>10.0.1.133</td><td>0100</td></tr><tr><td>6</td><td>PCU3</td><td>10.0.1.134</td><td>1100</td></tr><tr><td>6</td><td>PCU4</td><td>10.0.1.135</td><td>1011</td></tr><tr><td>8</td><td>PCU1</td><td>10.0.1.136</td><td>1010</td></tr><tr><td>8</td><td>PCU2</td><td>10.0.1.137</td><td>1001</td></tr></table></td><td>Derived Requirement</td><td>4S_05.-.Propulsion (Propulsion Control)</td><td>EVENT</td><td>N/A</td></tr>
</table>

MCM Enabling

Function Design

The MCM is enabled if traction shall be applied. The enable conditions are checked by TCMS according to train control parameters (master controller position, train speed).

Normal condition

The MCM shall be enabled only when the required conditions are present.
These required enable conditions are checked by TCMS which are then sent via PCU to enable both MCM.

The MCM shall be stopped in some circumstances.
These circumstances are checked by TCMS which will disable both MCM via PCU. For conditions when to enable and disable see below derived requirements.

Failure cases

Description of failure cases

1. Communication fault between TCMS and PCU. The MCMs connected to that PCU cannot be enabled thus no traction is available from these MCMs.
2. The MCM does not respond to the enable command. Traction will not be available for that MCM.
3. The MCM is enabled when not ordered. The motor will be magnetized even though the train is not intended to request traction. No impact on the operation.

Failure detection

Description of failure detection

1. The communication between CCU and the PCU is supervised by the CCU which will trigger an event stored in the diagnostic database.
2. PCU will generate an event that will be displayed on the IDU indicating that the MCM do not start on order.
3. PCU will generate an event that will be displayed on the IDU indicating that the MCM is started without an order.

Failure handling

Description of failure handling

1. Fault will be logged in the diagnostic database and checked at maintenance. Unscheduled workshop visit at the earliest after two days.
2. No action defined for the driver. Fault will be checked during maintenance. To be maintained at next scheduled workshop visit.
3. No action defined for the driver. Fault will be checked during maintenance. To be maintained at next scheduled workshop visit.

Requirements

ID	2F_05.01.Zefiro-Europe.TRS.163
Description	The MCM shall be stopped when – traction safe trainline is low OR – Train speed is 0 and Master Controller is not in a traction position – doors are released
Safety level	N/A

ID	2F_05.01.Zefiro-Europe.TRS.89
Description	MCM shall be enabled when the Master Controller is in traction position or train speed is >0.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_05.01.Zefiro-Europe.CONCEPT.115	Each individual MCM in the train (single or multiple) shall be enabled (started) when the Traction Controller is in a traction position OR Train speed is greater than 0 OR Emergency mode is active. The individual MCM shall be disabled (stopped) when – Traction safe trainline is de-energised OR – Train is towed OR – Train speed is 0 and Traction Controller has not been in traction position for at least 2 seconds OR – "Traction drive major mechanical fault" is set OR – MCM is cut out OR – The doors are ordered to release. If both enable and disable conditions are true at the same time the MCMs shall be disabled.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_05.01.Zefiro-Europe.CONCEPT.204	The PCU shall enable the MCM if: – signal MCM enable received via IP	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.622	If the MCM is not started (enabled) when ordered by TCMS then an event shall be generated.	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)	EVENT	N/A
2F_05.01.Zefiro-Europe.CONCEPT.623	If the MCM is started (enabled) when **not** ordered by TCMS then an event shall be generated.	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)	EVENT	N/A

LCM Interlacing

Function Design

The phase shift set point is calculated for each line converter in the train according to the number of total active line converters in the train and their relative location. The aim is to minimize the harmonics generated out on the supply line by distributing the phase shift evenly for all active line converters in the train.
The location of each line converter is calculated according to total number of units in the train, unit position in the train, number of active LCMs in the unit and position of the active line converter in respectively unit.
One unit corresponds to one 4-car segment, which implies the train consists of unit 1-and 2. Each unit includes four LCMs 1 to-4 divided into two propulsion systems. In case of one LCM is faulty/cut-out all LCM positions will be recalculated. In multiple operations the train consists of unit 1 to-4 and are divided into four propulsion systems.

Normal condition

The train function of interlacing is done by TCMS by notifying the PCU of the position of LCM and the amount of active LCM in the train. A switching angle shall be assigned to every individual LCM.
Though the switching angles of the LCM on the train will be evenly distributed over 360 degrees. This train function of interlacing is achieved by TCMS by notifying the PCU of the position of LCM and the amount of active LCM in the train.

The PCU shall calculate the corresponding phase shift set point for each individual LCM regarding the position of the LCM and the amount of active LCM in the train with the following criteria:

Normal mode
All LCM's in operation in one train unit is symmetrical interlaced which means that the LCM's for one train unit are interlaced with an angle of 45° if all four LCM's are in operation, i.e. 0°, 45°, 90° and 135° (The switching between the two 4QCs in one LCM shall be interlaced with an angle of 180°).
Additional offset angles shall be used for each transformer:
If four transformers are connected on the train the offset shall be 11.25°, i.e 0°, 11.25°, 22.5°, and 33.75°.

Degraded modes
The LCM's for one train unit are interlaced with an angle of 60° if only three LCM's are in operation, i.e. 0°, 60° and 120°
the LCM's for one train unit are interlaced with an angle of 90° if only two LCM's are in operation, 0° and 90°.
It is not allowed to have only one LCM in operation
If three transformers are connected on the train the offset shall be 15°, i.e 0°, 15°, 30°.
If two transformers are connected on the train the offset shall be 22,5°, i.e. 0° and 22,5°.

Failure cases

Description of failure cases

1. Incorrect position of active LCM in a unit is sent to the PCU. Wrong phase shift set point will be calculated by the PCU. The EMC limit could be exceeded.
2. Incorrect number of active LCMs in unit sent to PCU. Wrong phase shift set point will be calculated by the PCU. The EMC limit could be exceeded.
3. Incorrect unit position in train sent to PCU. Wrong phase shift set point will be calculated by the PCU. The EMC limit could be exceeded.
4. Incorrect total number of units in train sent to the PCU. Wrong phase shift set point will be calculated by the PCU. The EMC limit could be exceeded.
5. Communication fault between TCMS and PCU. The LCM in affected PCU will not be active.

Failure detection

Description of failure detection

1. Not detected.
2. Not detected.
3. Not detected.
4. Not detected.
5. The communication between CCU and the PCU is supervised by the CCU which will trigger an event stored in the diagnostic database.

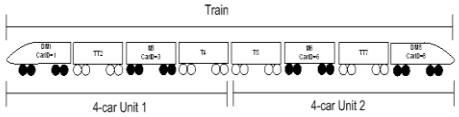
Failure handling

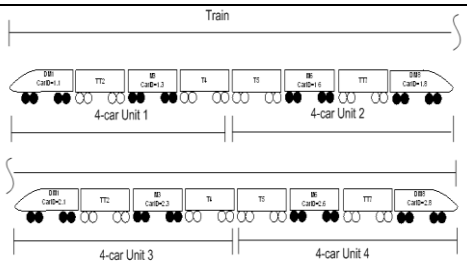
Description of failure handling

1. No failure handling.
2. No failure handling.
3. No failure handling.
4. No failure handling.
5. The number of LCMs in unit and the position of LCM in unit will be re-calculated by TCMS and sent to the PCU. The PCU will then re-calculate the phase shift set point for remaining LCMs in the unit. Unscheduled workshop visit at the earliest after two days.

Requirements

ID	2F_05.01.Zefiro-Europe.TRS.191
Description	Interlacing between line converters shall be provided: the line converters shall switch with shifted pulse pattern between each other. The switching angles shall be distributed evently over the whole train over 360 degrees.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_05.01.Zefiro-Europe.CONCEPT.239	The following signals shall be calculated and sent to each PCU in the train: – Position of active LCM in unit (half-train) – Number of active LCMs in unit (half-train) – Unit position in Train – Total number of units in train The LCM shall be considered active if the status from the LCM is not faulty and the LCM is not cut-out from the IDU and communication is OK between TCMS and the PCU. In single train operation the above signals shall be calculated based on following picture and table:  **Car identity / Total number of Units in Train / Unit position in Train / Number of active LCMs in unit / Position of LCM in unit** 1 / 2 / 1 / 4 / 1 1 / 2 / 1 / 4 / 3 3 / 2 / 1 / 4 / 2 3 / 2 / 1 / 4 / 4 6 / 2 / 2 / 4 / 4 6 / 2 / 2 / 4 / 2 8 / 2 / 2 / 4 / 3 8 / 2 / 2 / 4 / 1 In multiple train operation the above signals shall be calculated based on following picture and table:	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

	 **Car identity / Total number of Units in Train / Unit position in Train / Number of active LCMs in unit / Position of LCM in unit** 1.1 / 4 / 1 / 4 / 1 1.1 / 4 / 1 / 4 / 3 1.3 / 4 / 1 / 4 / 2 1.3 / 4 / 1 / 4 / 4 1.6 / 4 / 2 / 4 / 4 1.6 / 4 / 2 / 4 / 2 1.8 / 4 / 2 / 4 / 3 1.8 / 4 / 2 / 4 / 1 2.1 / 4 / 3 / 4 / 1 2.1 / 4 / 3 / 4 / 3 2.3 / 4 / 3 / 4 / 2 2.3 / 4 / 3 / 4 / 4 2.6 / 4 / 4 / 4 / 4 2.6 / 4 / 4 / 4 / 2 2.8 / 4 / 4 / 4 / 3 2.8 / 4 / 4 / 4 / 1				
2F_05.01.Zefiro-Europe.CONCEPT.398	For each LCM that is not in operation the number of LCMs in unit shall be decreased by 1 and all LCM positions with a higher position than the non-available LCM (within the same unit) shall be decreased by 1.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0
2F_05.01.Zefiro-Europe.CONCEPT.659	For each high voltage unit that is faulty (transformator is isolated) the Number of units shall be decreased by 1 and Unit position for units with a higher position than the non-available unit number shall be decreased by 1.	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_05.01.Zefiro-Europe.CONCEPT.252	The PCU shall indicate to TCMS if the LCM is faulty.	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.285	PCU shall calculate the switching angle for individual LCM depending on the train wide LCM status report from TCMS.	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.667	If only one LCM is in operation on one transformer at normal operation (not startup or short blocking conditions) then the LCM shall be stopped and an event shall be set indicating that the LCM is stopped due to EMC restrictions.	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		

Brake Backup mode

Function Design

Backup braking is a propulsion system function which is used to feed the auxiliary system and regulate the DC link if no line voltage is available. A small electro dynamical braking force is generated by the traction motors to feed energy into the DC link. Charging of the DC-link from the traction motors will also be available when the train is towed by using the pre-charge unit.

Normal condition

If the PCU received from TCMS the towing signal or the crossing neutral sections signal, or system separation section the MCM will be working in backup braking mode and will feed the DC link.

Failure cases

Description of failure cases

1. Communication fault between TCMS and PCU. Brake back-up mode will not be available.
2. No DC-link power supply due to faulty pre-charge unit.
3. Requested signal not sent to the PCU. The DC-link will not be charged.

Failure detection

Description of failure detection

1. The communication between CCU and the PCU is supervised by the CCU which will trigger an event stored in the diagnostic database.
2. PCU supervises the pre-charging unit. No DC link voltage or missing feedback from the pre-charging unit.
3. Probably the towing switch on the driver's desk is faulty. The towing switch is not in the scope of Drive Control SFRS.

Failure handling

Description of failure handling

1. Fault will be logged in the diagnostic database and checked at maintenance. Unscheduled workshop visit at the earliest after two days.
2. No backup braking for that Propulsion System. To be maintained at next scheduled workshop visit. An event shall be sent in case of failure.
3. Unknown.

Requirements

ID	2F_05.01.Zefiro-Europe.TRS.151
Description	The MCM shall work in Brake Backup mode to supply auxiliary power and to regulate DC link voltage in the following cases: – Crossing neutral sections – Loss of line voltage – Towing operation
Safety level	N/A

Backup braking is used for the propulsion system to feed and regulate the DC link if no line voltage is available. A small electro dynamical braking force is generated by the traction motors to feed energy into the DC link.

Nr	Description	Type	Derived to	User Interface	SIL
2F_05.01.Zefiro-Europe.CONCEPT.130	The MCM shall work in Brake Backup mode to supply auxiliary power and to regulate DC link voltage in the following cases: – Crossing neutral sections (status provided by TCMS over IP) – Crossing System separation Section(status provided by TCMS over IP) – Towing operation (status provided by TCMS over IP)	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.210	TCMS shall provide the following status signals to PCU: – towing operation active, this train is "towed" – neutral section passage is active – System separation Section passage is active	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

Boost Mode

Function Design

Boost mode is a propulsion system function used to compensate the reduction of traction power of one part of the train by using extra reserves from the remaining traction units.

Normal condition

TCMS will distribute the train status data dependent on the number of active MCM in the train.

The Boost mode shall be enabled when some traction power is lost, the remaining traction units shall be used to compensate this loss as much as possible.

Failure cases

Description of failure cases

A) Number of operational MCMs in train is falsely calculated by TCMS. The boost mode will falsely be activated or not activated when requested depending on the number of operational MCMs.
B) Communication between TCMS and PCU is faulty. The other PCUs will compensate for the lost traction in the Propulsion package

Failure detection

Description of failure detection

A) None identified.
B) The communication between CCU and the PCU is supervised by the CCU which will trigger an event stored in the diagnostic database.

Failure handling

Description of failure handling

A) None.
B) Fault will be logged in the diagnostic database and checked at maintenance. Unscheduled workshop visit at the earliest after two days.

Requirements

ID	2F_05.01.Zefiro-Europe.TRS.74
Description	In case of loss of traction power the remaining traction units shall be used to compensate the reduction as much as possible. If the remaining traction units have sufficient thermal reserves under the current external temperature conditions, the lost traction power shall be compensated.
Safety level	N/A

Nr	**Description**	**Type**	**Derived to**	**User Interface**	**SIL**
2F_05.01.Zefiro-Europe.CONCEPT.135	In case of any loss of traction power on the train, the remaining traction units shall be used to compensate the reduction as much as possible. If the remaining traction units have sufficient thermal reserves under the current external temperature conditions, the lost traction power shall be compensated. The compensation of traction power shall be done on two levels on command from TCMS.	Derived Requirement	4S_05.-.Propulsion		N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR	In 3kV mode the Boost mode is enabled if the propulsion components Temperature is lower of the followinr thresholds: CCM heat sink temperature < 70°C CCM internal air temperature < 65°C		4S_05.-.Propulsion (Propuslion Control)		

	MCM heat sink temperature < 65°C MCM internal air temperature < 65°C motor temperature < 200°C chopper inductor temperature < 130°C line inductor temperature < 155°C				
2F_05.01.Zefiro-Europe.CONCEPT.HR	In 25kV mode the Boost mode is enabled if the propulsion components Temperature is lower of the followinr thresholds: LCM heat sink temperature < 70°C LCM internal air temperature < 65°C MCM heat sink temperature < 65°C MCM internal air temperature < 65°C motor temperature < 200°C Primary damp link inductor temperature < 140°C 2nd harmonic inductor temperature < 70°C Transformer Temperature < 115°C		4S_05.-.Propulsion (Propuslion Control)		
2F_05.01.Zefiro-Europe.CONCEPT.HR	if any temperature is above the maximum admissible the time before activating boost mode after temperature is back to normal has > 1800sec		4S_05.-.Propulsion (Propuslion Control)		
2F_05.01.Zefiro-Europe.CONCEPT.666	Boost mode shall only be set at the following conditions – Ambient temperature is less or equal to 25degC. – Selected system is 25kV AC or 3 kV DC. – Line voltage is greater or equal to 3.4 kV DC at 3 KV DC system to activate boost mode. Note: This is only a start condition, boost mode shall not be deactivated due to line voltage below 3.4 kV DC.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_05.01.Zefiro-Europe.CONCEPT.286	"Boost mode 1" shall be set to all PCUs when 1, 2 or 3 MCMs in the train are not available.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_05.01.Zefiro-Europe.CONCEPT.665	"Boost mode 2" shall be set to all PCUs when 4 MCMs in the train are not available.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

Slip Slide Control

Function Design

The wheel slide protection and wheel slip protection are used to detect and prevent the wheel-slip during braking and wheel-slide during acceleration by regulating the traction torque depending on the adhesion.

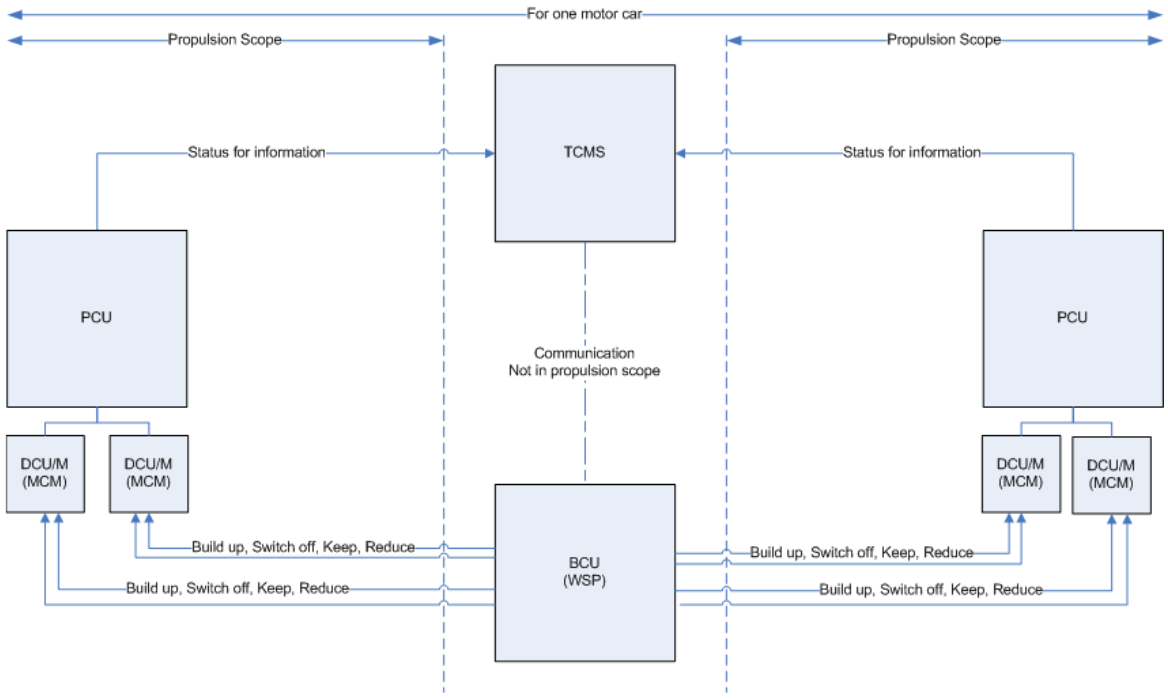
Normal condition

This function is based on the communication between Braking System, Propulsion System and TCMS. The WSP function in the braking system is managed by WSP control unit but if a WSP control unit is faulty the BCU realizes this function.

Two wired signals are sent by WSP and read by each MCM. The status of these signals are also sent to TCMS.

One wired signal "electro dynamic (ED) brake off" shall be sent by WSP and read by each MCM. The statuses of these signals are also sent to TCMS.

TCMS shall send the train speed to PCU. The signal shall be calibrated considering the wheel diameters.



Failure cases

Description of failure cases
The WSP control is managed by Braking system. No failure cases will be defined in this document.

Failure detection

Description of failure detection
N/A

Failure handling

Description of failure handling
N/A

Requirements

ID	2F_05.01.Zefiro-Europe.TRS.161
Description	The traction axles shall be equipped with an anti-slip system.
Safety level	N/A

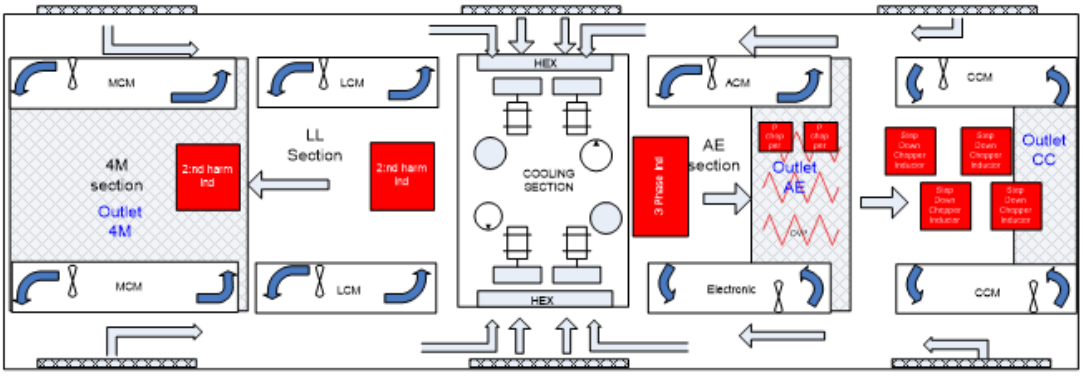
ID	2F_05.01.Zefiro-Europe.TRS.201
Description	The traction axles shall be equipped with an anti-slide system, this shall interact with the brake systems wheel side protection (WSP).
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_05.01.Zefiro-Europe.CONCEPT.265	The WSP shall provide two wired signals to PCU (DCU/M) for slide control management with the following codes: – build up – switch off – keep – reduce the status of the signals shall be fed back to TCMS for information.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.01.Zefiro-Europe.CONCEPT.268	PCU shall read two wired signals for slide control management with the following codes: – build up , – switch off – keep – reduce the status of the signals shall be fed back to TCMS for information. Command / WSP_Hold / WSP_Vent Build-up ED brake force: 0 / 0 Keep ED brake force: 1 / 0 Reduce ED brake force: 1 / 1 Switch-off ED brake force: 0 / 1	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR02	Each MCM detects the slipping condition and implements the anti-slip control by torque management	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		SIL 0
2F_05.01.Zefiro-Europe.CONCEPT.430	If Emergency brake is active then the PCU shall be notified.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

Traction Converter Cooling

Function Design

The converter cooling system consists of internal section fans and an external cooling system situated in the middle of converter and composed of two indipendent single modules, one for each half converter.
The internal sections fans work at a fix rotation speed fix



Each section of the external cooling system has a dedicated pump to control the cooling fluid and two fans working a two speed. Each PCU handles a section of the external cooling, starting the water pump and controlling the fans speed.
The pump is started when the medium voltage is available.
The fans speed depends on the components temperature and on the on the vehicle speed.
In detail the temperature taken in account are the heat sink and internal air temperature of each module, the second harmonic inductor temperature acquired by MOCU/L and situated in LL and 4Msections, the step down inductors temperature, situated in CC section, the three phase transformer, situated in AE section. The MOCUs send the components temperatures acquired to the PCUt

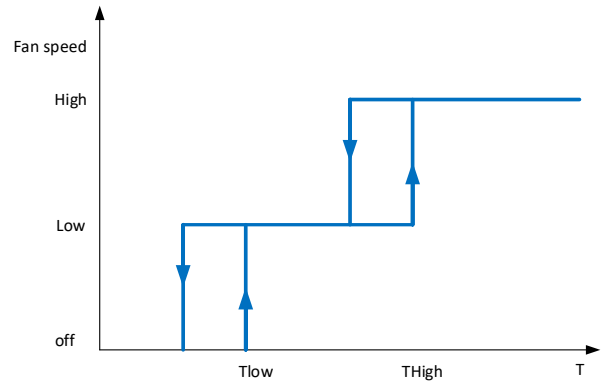


Figure 2 external fans speed regulation

The fan speed regulation law is shown in the Figure 2 but the temperature thresholds depend on the component temperature and the vehicle speed as reported in the following tables
The fan and pump status is monitored, checking the coherence between order and activation status
Also the cooling water level is monitored

Internal air Temperature (CCM,MCM,LCM,ACM)		
	T [°C)	
Level	v<10 km/h	v>20km/h
TLowRst	na	na
Tlow	na	na
THighRst	60	55
Thigh	63	58

Heat sink Temperature (CCM,MCM,LCM,ACM)		
	T [°C)	
Level	v<10 km/h	v>20km/h
TLowRst	0	0
TLow	5	5
THighRst	60	45
Thigh	65	50

3-phase Transformer (MOCU/A) step down inductors (MOCU/C)		
	T [°C)	
Level	v<10 km/h	v>20km/h
TLowRst	0	0
TLow	10	10
THighRst	105	65
Thigh	110	70

harmonic inductor (MOCU/L)		
	T [°C)	
Level	v<10 km/h	v>20km/h
TLowRst	0	0
TLow	10	10
THighRst	65	50
Thigh	68	53

Requirements

Nr	Description	Type	Derived to	User Interface	SIL
2F_05.01.Zefiro-Europe.CONCEPT.HR18	The pump activation is started by PCU digital output and its activation status is acquired by PCU digital input	Derived Requirement	4S_05-.Propulsion		N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR19	The pump is started when the medium voltage is available	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR36	The level status is acquired by PCU digital input	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR20	The fan speeds selection is started by PCU digital output and the selected speed status is acquired by PCU digital input	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR21	The fans speed depends on the component temperature and on the on the vehicle speed	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR22	Each MOCU acquires module heat sink and internal air temperature and send it to PCU	Derived Requirement	4S_05.-.Propulsion		N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR23	The MOCU/L acquire the second harmonic inductor temperature and send it to PCU	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR24	The MOCU/C acquire the step down inductors temperatures and send them to the PCUt	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A

2F_05.01.Zefiro-Europe.CONCEPT.HR25	The MOCU/A acquires the three phase transformertemperature and send it to PCU	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR26	Each PCU actives the fan at low speed if at least a component temperature of its converter section is higher than TLow (level depending on the component and vehicle speed)	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR27	Each PCU switches off the fan if all the component temperatures of its converter section are lower than TLowRst (level depending on the component and vehicle speed)	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR28	Each PCU actives the fan at high speed if at least a component temperature of its converter section is higher than THigh (level depending on the component and vehicle speed)	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR29	Each PCU switches the fan speed from high to low if all the component temperatures of its converter section are lower than THighRst (level depending on the component and vehicle speed)	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR30	If fan low speed doesn't active on order high speed shall permanently override low speed order and Event 'External fan low speed fault' **shall** be indicated	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR31	If fan high speed doesn't active on order low speed shall permanently override high speed order and the performnce shall be reduced at 50% in that converter section Event 'External fan high speed fault' **shall** be indicated	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR32	If acknowledge is not received either from low and high speed, high speed **shall** permanently override low speed order and the the performance shall be reduced at 50% in that converter section Event 'External fan high and low speed fault' shall be indicated	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR33	If fan low speed doesn't switch off on order, low speed shall permanently override high speed order and the performance shall be reduced at 50% in that converter section and the event 'External fan low speed fault' **shall** be indicated	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR34	If fan high speed doesn't switch off on order, high speed shall permanently override low speed order and the Event 'External fan high speed fault' **shall** be indicated	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR35	If the pump is activated and water pump cosfi is low and the water tank level is not ok. The event "*Converter box low water tank level*" is created and the pump is switched off and the converter is soft switched off. The event shall be set permanent if the failure is detected twice (2) within 30 minutes.	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR37	If the pump is activated and the pump contactor is closed and water pump cosfi is low and the water tank level is ok. The event "*Converter box water pump cosphi relay fault "* event is created and the pump is switched off and the converter is soft switched off. The event shall be set permanent if the failure is detected twice (2) within 30 minutes.	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR38	If the pump is activated but the pump contactor doesn't close and water pump cosfi is high. The Event '*Converter box water pump contactor ON sequence fault*' shall be indicated.	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR39	If the pump is de-activated but the pump contactor remain close. The Event '*Converter box water pump contactor OFF sequence fault*' is created.	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A

2F_05.01.Zefiro-Europe.CONCEPT.HR56	If the pump is activated and the pump contactor is not closed and water pump cosfi is low. The event '*Converter box water pump error activation fault"* is created and the pump is switched off and the converter is soft switched off. The event shall be set permanent if the failure is detected twice (2) within 30 minutes.	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A

Traction Converter Temperatures supervision

Function Design

The converter component temperature supervisione checks if the temperatures are in an admissible range to avoid component damnage. For each componet are defined different opertive temperature level are defined. A low temperature, a high temperature level, a HighMax level. When the component reaches the High temperature the performance are reduced, when the converter reaches the High Max temperture the converter is blocked. The thermal protection reset is possible when the temperature becomes lower than TLow
The tables below report the levels Low, High and High Max for each component ant the action requested.

Internal Air Temperature					
	ACM	CCM	LCM	MCM	
ThLow	-	65	65	65	Performance Reduction Reset
THigh	-	68	68	68	Performance Reduction
ThighMax	70	70	70	70	Converter blocked

Heat sink Temperature					
	ACM	CCM	LCM	MCM	
ThLow	-			65	Performance Reduction Reset
THigh	-	72	72	70	Performance Reduction
ThighMax	76	75	75	75	Converter blocked

2° Harm Inductor Temperature (LCM)		
	T [°C)	
ThLow	70	Performance Reduction Reset
Thigh	79	Performance Reduction
ThighMax	83	Converter blocked (AC)

Phase Inductor Temperature (CCM)		
	T [°C)	
ThLow	130	
THigh	140	7200 sec Max
ThighMax	160	Converter blocked

3-pfase TransformerTemperature (ACM)		
	T [°C)	
TLow	100	
ThighMax	122	Converter blocked

Requirements

Nr	Description	Type	Derived to	User Interface	SIL
2F_05.01.Zefiro-Europe.CONCEPT.HR40	if the internal air temperature or heat sink temperature of MCM or CCM or LCM is above the High level the performance shall be reduced of 50% the event Module xx temperature is High is created	Derived Requirement	4S_05-.Propulsion		N/A

2F_05.01.Zefiro-Europe.CONCEPT.HR41	if a module internal air temperature or heat sink temperature is above the HighMax level OR the temperature is above Thigh level for 3600sec the converter shall be soft blocked the event Module xx temperature is Too High is created	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR42	The reset module thermal protection is possible when the temperature is below the Low temperature	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR43	if the AC voltage system is selected and 2° Harm Inductor Temperature is above the High level the performance reduced of 50% the event 2° Harm Inductor Temperature is High is created	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR44	if the AC voltage system is selected and 2° Harm Inductor Temperature is above the HighMax level OR the temperature is above High level for 3600 sec the the converter shall be soft blocked the event 2° Harm Inductor Temperature is too High is created	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR45	if the DC voltage system is selected and Chopper phase Inductor Temperature is above the High level the performance shall be reduced of 50% the event chopper phase Inductor Temperature is High is created	Derived Requirement	4S_05.-.Propulsion		N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR46	if the DC voltage system is selected and Chopper phase Inductor Temperature is above is above the HighMax level OR if the temperature is above the Thigh level for more than 7200seconds the converter shall be soft blocked the event Chopper phase Inductor Temperature is too High is created	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR47	if 3-pfase TransformerTemperature is above the HighMax level the ACM module shall be blocked the event 3-pfase TransformerTemperature is too High is created	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR48	The component thermal protection are resettable if the component temperature becomes lower than the Low temperature	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A

Traction Motor Cooling

Function Design

For the cooling of the traction motor there is one traction motor fan for each traction motor.
Each fan is controlled by a variable frequency converter. The set point for the converter is set by 2 analog output of each PCU. one for each motor of the bogie, the start of the converter is from TCMS digital output.
Starting from 0 speed, when the vehicle speed is lower than minimum speed, minSpeedxMotFan, (9km/h) the converter frequency depends only on the Motor temperature as reported below, respecting the valid thresholds in this speed range.
When the vehicle speed is greater than minimum speed, minSpeedxMotFan, (9km/h) depends also on the rate reference requested. If the rate reference is lower than RateRefFull (80%) the converter frequency depends only on the Motor temperature respecting the valid thresholds in this speed range, if the requested effort greater then RateRefFull the converter frequency is the maximum.
If the speed reduction from a speed greater then minSpeedxMotFan occurs an hystersys has to be applied to change the theresholds and a gradual change between the frequency reference. The frequency percent change has not to be faster than Df/Dt= 100%/ 5sec.

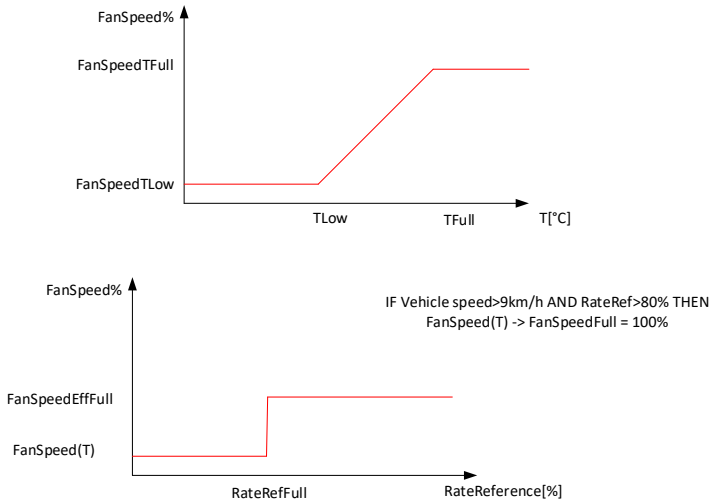


Figure 3 Traction Motor Fan control

	Vel>minSpeedxMotFan	Vel< minSpeedxMotFan
FanspeedTFull	100%	100%
FanspeedTLow	50%	10%
Tlow (°C)	110	80
Tfull (°C)	150	150
FanSpeedEffFull	100%	-

The frequency converter is supervised via digital interface to a RIO-unit.

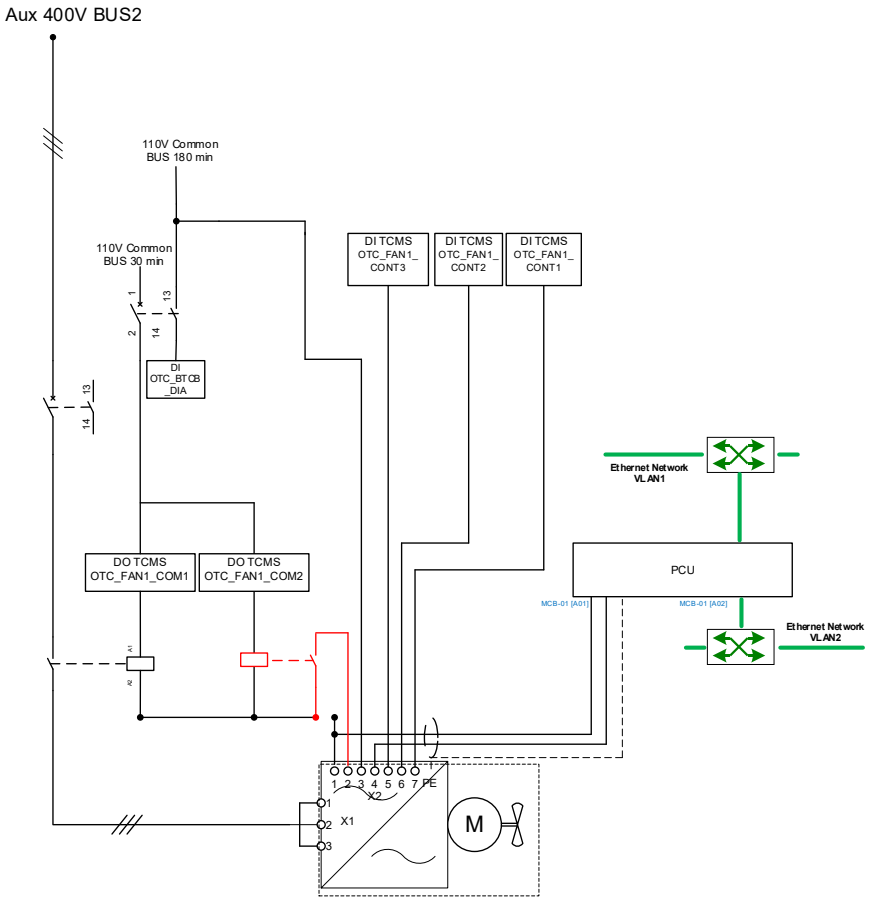


Figure 4 traction motor fans regulation

Normal condition

In normal condition the PCU is controlling the fan via the frequency converter.

Failure cases

Description of failure cases

1. Fan fault
2. Frequency converter fault
3. PCU interface fault

Failure detection

Description of failure detection

A) By temperature supervision of motor.
B) Through status bits from frequency converter.
C) PCU unit supervision or Fan is not starting on order

Failure handling

Description of failure handling

A) Motor converter will be blocked.
B) Fault will be logged in the diagnostic database and checked at maintenance.
C) Fault will be logged in the diagnostic database and checked at maintenance.

Requirements

Nr	Description	Type	Derived to	User Interface	SIL
2F_05.01.Zefiro-Europe.CONCEPT.649	The converter frequency shall be set from PCU to the frequency converter for traction motor cooling.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR03	The motor cooling converter frequency shall be set from PCU to the frequency converter for traction motor cooling dependign on motor temperature, requested effort and vehicle speed.	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR04	When the medium voltage is available the motor cooling converter starting is requested via IP bus.	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR05	The motor cooling converter frequency shall be set to a minimum value (10 Hz) when the motor cooling converter starting is requested to TCMS.	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR06	Starting from 0 speed, when the vehicle speed is lower than a minimum speed, minSpeedxMotFan, (9km/h) the converter frequency depends only on the Motor temperature. It increases linearly from 10% to 100% when the temperature increase from 80°C to 150°C	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR07	When the vehicle speed is greater than minimum speed, minSpeedxMotFan, the converter frequency depends on the rate reference requested. If the rate reference is lower then RateRefFull (80%) the converter frequency depends only on the Motor temperature: It increases linearly from 50% to 100% when the temperature increase from 110°C to 150°C if the requested effort is greater than RateRefFull the converter frequency shall be set to the maximum.	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR08	If a vehicle speed reduction from a speed greater than minSpeedxMotFan occurs an hystersys has to be applied to change the theresholds	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR09	The frequency percent change has not to be faster than Df/Dt= 100%/ 5sec.	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.650	The "start code" to the frequency converter shall be set when ordered from TCMS DO.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR10	The "start code" to the frequency converter shall be set when the request is received from PCU	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_05.01.Zefiro-Europe.CONCEPT.651	The status bit "MK1" from the frequency converter shall be connected to TCMS DI.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_05.01.Zefiro-Europe.CONCEPT.652	The status bit "MK2" from the frequency converter shall be connected to TCMS DI.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_05.01.Zefiro-Europe.CONCEPT.654	The status bit "MK3" from the frequency converter shall be connected to TCMS DI.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_05.01.Zefiro-Europe.CONCEPT.673	The 3 status bits from frequency converter shall be evaluated to receive the status of the converter.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

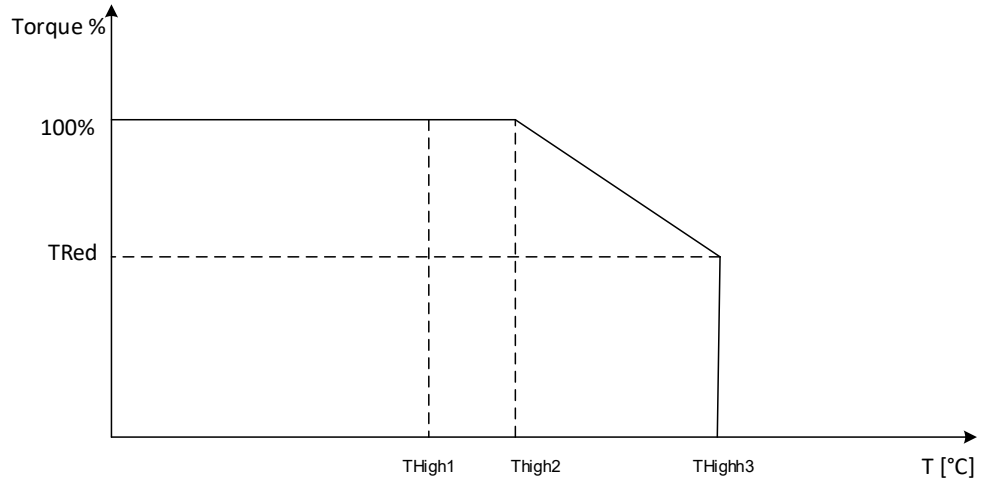
2F_05.01.Zefiro-Europe.CONCEPT.655	The "Status Traction motor cooling inverter OK" shall be set to PCU when the frequency converter is not permanently faulty and 400V supply is available for the traction motor cooling fan. Motor Fan OK signal shall be removed in 20 sec, If 400V Missing OR Permanent Fault is active [Mk1=0; Mk2=0; Mk3=0] OR Temporary Fault is active [Mk1=0; Mk2=1; Mk3=1] OR (DBC_ACO_C_ActColMt = 1 (Command Signal From PPC) AND DBC_ACO_S_MK1 = 0 (bit MK1) for more than 10 sec with 400V present and contactor closed (to avoid start-up misalignements))	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_05.01.Zefiro-Europe.CONCEPT.680	Any time TMF does not start on order, TCMS shall try a new start up resetting "start bit" command by DO. After 3 consecutive missing starts are performed within 60 seconds, an event shall be set and related MCM shall be blocked. Fault count shall be reset anytime 400Vac power supply is reset	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_05.01.Zefiro-Europe.CONCEPT.671	When the status from TCMS "Traction motor cooling inverter ok" is not set the motor converter shall be blocked. Note that the motor converter shall not be blocked due to inverter not ok when motor converter is in backup braking mode, e.g. during towing.	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.656	An event shall be set when the converter reports a failure according to the status bits. Permanent Fault shall be reset every time 400 V is not availeble on the consist Permanent fault iis defined if 400 V is provided to the TMF and the bit are all 0 [Mk1=0; Mk2=0; Mk3=0]	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.01.Zefiro-Europe.CONCEPT.672	An event shall be set when the converter does not start on order, i.e. when indication that "motor is working" is not set from the frequency converter when "start code" is set to the frequency converter.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.01.Zefiro-Europe.CONCEPT.670	The command to activate the cooling fan shall be set to TCMS when the reference frequency for the cooling fan is higher than 0.	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.657	The start order "start code" to the converter shall be set when requested from PCU and 3AC400V contactor "KTMC" has been set for at least 5s per each Motor car.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_05.01.Zefiro-Europe.CONCEPT.668	The 3AC400V contactor "KTMC" shall be closed when ordered from TCMS DO.	Derived Requirement	4S_09.02.09.-.Conventional Train Control – Circuits		N/A

2F_05.01.Zefiro-Europe.CONCEPT.669	oThe DO for closing of 3AC400V contactor "KTMC" of the traction motor cooling shall be set when 400V supply is available and a specific delay, per each Motor car, is elapsed Car / Delay (s) DM1 / 2 M3 / 4 M6 / 6 DM8 / 8 If the motor fan is in permanent fault for more than 60 sec the related DO for closing of 3AC400V contactor "KTMC" of the traction motor cooling shall be opened. After 60 sec. the DO for closing of 3AC400V contactor "KTMC" of the traction motor cooling shall be closed and an even shall be set “First attempt to restarting motor fan after a permanent fault “. If the motor fan is still in permanent fault after 60 sec. the related DO for closing of 3AC400V contactor "KTMC" of the traction motor cooling shall be opened. After 120 sec. the DO for closing of 3AC400V contactor "KTMC" of the traction motor cooling shall be closed and an even shall be set “Second attempt to restarting motor fan after a permanent fault “. If the motor fan is still in permanent fault after 60 sec. the related DO for closing of 3AC400V contactor "KTMC" of the traction motor cooling shall be opened. After 300 sec. the DO for closing of 3AC400V contactor "KTMC" of the traction motor cooling shall be closed and an even shall be set “Third attempt tgo restarting motor fan after a permanent fault “. If after 60 sec. the motor fan is still in permanent fault the motor fan shall be blocked (no restart are allowed until permanent fault is set to zero) and an event shall be set “To many permanent fault of the motor fan”. If during this sequence the 400 V is not available, the sequence shall be stopped and the normal sequence shall be activated.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

Traction Motor Temperature Supervision

Function Design

Each motor is monitored by a temperature sensor connected to the Propulsion control (MOCU/M)
A perfromance reduction is realised if the motor temperature reaches a value greater than a first threshold of High Temperature (Thigh2), the torque will start to be linearly reduced and the status “High motor temp” is set for corresponding motor.
The torque reduction will stay active until the temperature decreases to a lower temperature (Thigh1).
The status “High motor temp” will be reset when the temperature reaches the Low motor temperature threshold (THigh1).
If the temperature remains above the High Temperature level for a specific time, 7200sec, or the Maximum temperature level (Thigh3) is exceeded an overtemperature fault is indicated and the traction inverter is soft blocked. In such case, the converter is reactivated when the temperature decreases to the Low temperature level



Where THigh1 = 200°C
THigh2 = 220°C
THigh 3= 230°C (torque 0%)
TMax = 250°C
TRed = 50% of maximum rate reference

Requirements

Nr	Description	Type	Derived to	User Interface	SIL
2F_05.01.Zefiro-Europe.CONCEPT.HR11	The motor temperature sensors shall be connected to MOCU/M. Each MOCU/M acquires the motor temperature sensor of the controlled motor	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR12	A perfromance reduction is realised if the motor temperature reaches a value greater than a first threshold of High Temperature (220°C), the torque will start to be linearly reduced and the status “High motor temp” is set for corresponding motor	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR13	The torque reduction will stay active until the temperature decreases to a Lower temperature (200°C).	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A

	The status “High motor temp” will be reset when the temperature reaches the Low motor temperature threshold				
2F_05.01.Zefiro-Europe.CONCEPT.HR14	If the temperature remains above the High Temperature level for a specific time, 7200sec, an overtemperature fault is indicated and the traction inverter is soft blocked. The converter is reactivated when the temperature decreases to the Low temperature level	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR15	If the temperature reaches the Maximum temperature level (230°C) an overtemperature fault is indicated and the traction inverter is soft blocked. The converter is reactivated when the temperature decreases to the Low temperature level	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR55	A Protection logic shall be implemnted if a temperature difference of 100°C is detecetd between two active motors	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR16	The motor temperature sensors are considered faulty if the sensor detects a temperature outside the limits -50..+530 °C. In this case an event is created	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.HR17	If the motor temperature sensor is faulty the torque is reduced at 400Nm in traction and – 400 Nm in braking	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A

Commissioning and Test

Activation of Transient Recorder

Function Design

In order to achieve some tests and commissioning, the transient recorders within the propulsion equipment in the train shall be activated centrally from one CCU.

Normal condition

All transient recorders from each PCU shall be activated by receiving a trigger signal from TCMS.

Failure cases

Description of failure cases
The configuration and activation of the transient recorder is usually done during advanced fault finding. Identifying failures of function is not applicable because a failure will not have any impact on the operation of the train nor affect normal maintenance.

Failure detection

Description of failure detection
N/A

Failure handling

Description of failure handling
N/A

Requirements

ID	2F_05.01.Zefiro-Europe.TRS.97
Description	For commissioning and tests it shall be possible to activate all transient recorders within the Propulsion equipment in the train by setting a parameter in the TCMS
Safety level	N/A

In order to achieve some tests and commissioning, the transient recorders within the propulsion equipment in the train shall be activated

Nr	Description	Type	Derived to	User Interface	SIL
2F_05.01.Zefiro-Europe.CONCEPT.140	Activation of all transient recorders (trace function) within the Propulsion equipment shall be performed by receiving a trigger signal on IP (via PTU directly connected).	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.211	A simple parameter shall be defined to set all propulsion related transient recorders to active.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

Cut Out of Converter

Function Design

During commissioning and tests, it shall be possible to cut out (disable) all converter modules by pressing a soft button on IDU. This function shall only be accessible for maintenance personal.

Normal condition

From the IDU of the maintenance personal, it shall be presented a menu for cut out of every converter module per motor car.

From a signal on IP, it shall be possible to cut out every converter module individually.

Failure cases

Description of failure cases

A) Communication fault between TCMS and PCU. The LCM, ACM, CCM and MCMs controlled by the PCU will not be possible to be cut-out from the IDU.
B) Converter is not cut-out when order by TCMS.
C) Converter is cut-out or static cut-out when no order is present from TCMS.

Failure detection

Description of failure detection

A) The communication between CCU and the PCU is supervised by the CCU which will trigger an event stored in the diagnostic database.
B) An event will be generated and displayed on the IDU indicating that the converter does not cut-out on order.
C) An event will be generated and displayed on the IDU indicating that the converter is cut-out even though no cut-out order is present.

Failure handling

Description of failure handling

A) Fault will be logged in the diagnostic database and checked at maintenance.
B) Fault will be checked at maintenance.
C) Fault will be checked at maintenance.

Requirements

ID	2F_05.01.Zefiro-Europe.TRS.91
Description	For commissioning and tests it shall be possible to cut-out each LCM, CCM,ACM and MCM (via IDU)
Safety level	N/A

All converter modules can be cut out (disabled) by pressing a soft button on IDU. This function shall only be accessible for maintenance personal.

Nr	Description	Type	Derived to	User Interface	SIL
2F_05.01.Zefiro-Europe.CONCEPT.145	The IDU shall provide a menu for cut out (disabling) of the following converter modules individually per motor car: – 4 MCM – 2 LCM – 2 CCM – 1 ACM	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_05.01.Zefiro-Europe.CONCEPT.212	It shall be possible to cut out all of the converter modules individually upon a request indicated by IP signals.	Derived Requirement	4S_05.-.Propulsion		N/A
2F_05.01.Zefiro-Europe.CONCEPT.624	If the converter does not respond to the cut-out order from TCMS then an event shall be generated.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.01.Zefiro-Europe.CONCEPT.625	If the converter is cut-out but no cut-out order is present from TCMS then an event shall be generated.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

Maintenance Tests

Automatic Test

Function Design

The automatic propulsion system tests shall be possible by triggering on IDU by the maintenance personal.

Normal condition

The automatic propulsion system test shall be triggered by a soft button on IDU by a maintainer.

The automatic test can be performed under the precondition that the train is at standstill. Then the following sequence shall be performed:

1. Apply full service brake in train
2. Set Traction Block
3. Send Automatic test command to all PCU units in the train
4. Collect test status from all PCU after the test
5. Reset Traction Block (note: only test condition)

Automatic test is activated and forwarded by IP signal from TCMS.

PCU shall perform an automatic test.

PCU shall report the test status.
The results of the automatic test shall be reported via TDS events.

Failure cases

Description of failure cases

1. Communication fault between TCMS and PCU. The automatic test of the Propulsion module can not be tested.
2. The automatic test is not started when requested by TCMS even though conditions for test were fulfilled.
3. The automatic test is started when not requested by TCMS.

Failure detection

Description of failure detection

1. The communication between CCU and the PCU is supervised by the CCU which will trigger an event stored in the diagnostic database.
2. The PCU will generate an event (part of the standard product) if the automatic test can not be started when requested.
3. TCMS will supervise the "test in progress" signals sent from the PCU. If the automatic test is started when not requested protective shutdown will be requested to the PCU.

Failure handling

Description of failure handling

1. Fault will be logged in the diagnostic database and checked during maintenance.
2. Fault will be logged in the diagnostic database and checked during maintenance.
3. No action from maintenance staff. It will be notified that the Propulsion module has been disabled.

Requirements

ID	2F_05.01.Zefiro-Europe.TRS.99
Description	It shall only be possible to start the Automatic test when – Train speed is zero AND – Maintainer logged in on IDU
Safety level	N/A

ID	2F_05.01.Zefiro-Europe.TRS.167
Description	When the Automatic test is activated the following sequence shall manage the test: 1. Apply full service brake in train (equivalent to Brake Position 7 of Master Controller) 2. Set Traction Block 3. Send Automatic test command to all propulsion control's in the train 4. Collect test status from all propulsion control's 5. Reset Traction Block (note: only test condition)
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_05.01.Zefiro-Europe.CONCEPT.150	When the Automatic test is activated as indicated by IP signal from TCMS, the PCU shall perform an automatic test and report the test status after finalisation of the tests to TCMS using process data signals. The automatic test shall not be started if the speed is greater than 0. The automatic test shall be aborted if the speed is greater than 0.	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)		N/A
2F_05.01.Zefiro-Europe.CONCEPT.289	The results of the automatic test shall be reported via TDS events. Events related to the automatic test are part of the standard product.	Derived Requirement	4S_05.-.Propulsion (Propulsion Control)	EVENT	N/A
2F_05.01.Zefiro-Europe.CONCEPT.213	The automatic propulsion system test shall be triggered by a soft button on IDU by a maintainer under the preconditions that the train is at standstill,. Then the following sequence shall be performed: 1. Apply full service brake in train 2. Set Traction Block 3. Send Automatic test command to all PCU units in the train 4. Wait for all PCUs to finish the test. 5. If all PCUs indicates test ok then test passed shall be shown on the IDU. If one or more PCUs indicates test failed then test failed shall be shown on the IDU 6. Reset Traction Block (note: only test condition) 7. Reset full service brake	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.01.Zefiro-Europe.CONCEPT.214	The IDU shall provide a soft button for automatic traction system test. This button shall only be accessible in maintenance mode (all pantograph down).	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_05.01.Zefiro-Europe.CONCEPT.628	If the PCU indicates that the automatic test is in progress but the test has not been requested from the IDU then protective shutdown shall be requested to the PCU. In this situation an event shall be generated to be displayed on the IDU for the driver.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.01.Zefiro-Europe.CONCEPT.678	The IDU shall provide a soft button for line trip test. This button shall only be accessible in maintenance mode,.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_05.01.Zefiro-Europe.CONCEPT.679	The results of the line trip test shall be reported via TDS events.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

Condition Based Maintenance

CBM test start

The CBM test shall be command by TCMS to propulsion if the following conditions are valid:

1. test is command by the maintainer via IDU
2. the train has to be not in parking mode

If more than 168 hours are passed from the last test, an event shall be set from TCMS to the maintainer

Test

To activate the test to PPC, TCMS has also to send the variables below:

CCU-O - > PCU	Signal	Cycle time	Description	Scale
CT1CX403	CI_TCMS_XU_Bat	512 ms	Battery voltage	10=1V
CT1CX4C1:8	CI_TCMS_VU_Bat	512 ms	Valid bit: Battery voltage	L
CT1CX3C4:3	CI_TCMS_C_CbmTst	64 ms	Command: CBM test	L

the variable CI_TCMS_C_CbmTst has to be at one during the complete test

To set the variable CI_TCMS_C_CbmTst, TCMS has also to supervise if the following condition are true:
1. Train at standstill
2. 3-phase bus active
3. DC-link are charged, all CMM or LCM are in operation

During the test the conditions
1. 3-phase bus active
2. DC-link are charged, all CMM or LCM are in operation

Must not be take in consideration to abort the test

Once the test is finished, PPC will set to zero the variable below

PCU - > CCU-O	Signal	Cycle time	Description	Scale
CX1CT4C2:14	CBM_S_CbmTst	512 ms	Status: CBM test ongoing	L

If the test is not passed, PPC will set an event PPC CBM_E_CbmTstFl, this event has to set to the maintainer.
Once the test is completed and not fault event is set, the test is completed, TCMS shall set an event for the maintainer, a massage under the virtual push button has to appear with test completed.
TCMS has to abord the test if after 10 s CI_TCMS_C_CbmTst is 1, CBM_S_CbmTst is still 0. TCMS has to send an event to the maintainer that the CBM test has not started, a massaga under the virtual push button has to appear with test not started.
TCMS has to abord the test if after 130 s CI_TCMS_C_CbmTst is 1, CBM_S_CbmTst is still 1. TCMS has to send an event to the maintainer that the CBM test takes to long time, a massage under the virtual push button has to appear with test to long time.
During the complete test phase a massage has to be set on the IDU test page, that traction is not allowed due to the CBM test

Signal Interface

I/O Interface

Supplier system (propulsion)

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_05.01.Zefiro-Europe.CONCEPT.431	WSP control signal 1 for axle 1	L	2F_05.01.Zefiro-Europe.CONCEPT.268
2F_05.01.Zefiro-Europe.CONCEPT.432	WSP control signal 2 for axle 1	L	2F_05.01.Zefiro-Europe.CONCEPT.268
2F_05.01.Zefiro-Europe.CONCEPT.433	WSP control signal 1 for axle 2	L	2F_05.01.Zefiro-Europe.CONCEPT.268
2F_05.01.Zefiro-Europe.CONCEPT.434	WSP control signal 2 for axle 2	L	2F_05.01.Zefiro-Europe.CONCEPT.268
2F_05.01.Zefiro-Europe.CONCEPT.435	WSP control signal 1 for axle 3	L	2F_05.01.Zefiro-Europe.CONCEPT.268
2F_05.01.Zefiro-Europe.CONCEPT.436	WSP control signal 2 for axle 3	L	2F_05.01.Zefiro-Europe.CONCEPT.268
2F_05.01.Zefiro-Europe.CONCEPT.437	WSP control signal 1 for axle 4	L	2F_05.01.Zefiro-Europe.CONCEPT.268
2F_05.01.Zefiro-Europe.CONCEPT.438	WSP control signal 2 for axle 4	L	2F_05.01.Zefiro-Europe.CONCEPT.268

Outputs:

Nr	Functional name	Type	Linked to requirement
	TMF1 frequency set point	AO	2F_05.01.Zefiro-Europe.CONCEPT.649
	TMF2 frequency set point	AO	2F_05.01.Zefiro-Europe.CONCEPT.649
	TMF3 frequency set point	AO	2F_05.01.Zefiro-Europe.CONCEPT.649
	TMF4 frequency set point	AO	2F_05.01.Zefiro-Europe.CONCEPT.649

Supplier system (brake)

Inputs:

Outputs:

Nr	Functional name	Type	Linked to requirement
2F_05.01.Zefiro-Europe.CONCEPT.439	WSP control signal 1 for axle 1	L	2F_05.01.Zefiro-Europe.CONCEPT.265
2F_05.01.Zefiro-Europe.CONCEPT.440	WSP control signal 2 for axle 1	L	2F_05.01.Zefiro-Europe.CONCEPT.265
2F_05.01.Zefiro-Europe.CONCEPT.441	WSP control signal 1 for axle 2	L	2F_05.01.Zefiro-Europe.CONCEPT.265
2F_05.01.Zefiro-Europe.CONCEPT.442	WSP control signal 2 for axle 2	L	2F_05.01.Zefiro-Europe.CONCEPT.265
2F_05.01.Zefiro-Europe.CONCEPT.443	WSP control signal 1 for axle 3	L	2F_05.01.Zefiro-Europe.CONCEPT.265

2F_05.01.Zefiro-Europe.CONCEPT.444	WSP control signal 2 for axle 3	L	2F_05.01.Zefiro-Europe.CONCEPT.265
2F_05.01.Zefiro-Europe.CONCEPT.445	WSP control signal 1 for axle 4	L	2F_05.01.Zefiro-Europe.CONCEPT.265
2F_05.01.Zefiro-Europe.CONCEPT.446	WSP control signal 2 for axle 4	L	2F_05.01.Zefiro-Europe.CONCEPT.265
	AC System Detection Test		
	DC System Detection Test		

Supplier system (drivers desk)

Inputs:

Outputs:

TCMS SW

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_05.01.Zefiro-Europe.CONCEPT.661	Status bit MK1		2F_05.01.Zefiro-Europe.CONCEPT.655 2F_05.01.Zefiro-Europe.CONCEPT.673
2F_05.01.Zefiro-Europe.CONCEPT.662	Status bit MK2		2F_05.01.Zefiro-Europe.CONCEPT.656 2F_05.01.Zefiro-Europe.CONCEPT.673
2F_05.01.Zefiro-Europe.CONCEPT.663	Status bit MK3		2F_05.01.Zefiro-Europe.CONCEPT.651 2F_05.01.Zefiro-Europe.CONCEPT.673

Outputs:

Nr	Functional name	Type	Linked to requirement
2F_05.01.Zefiro-Europe.CONCEPT.664	Start code (traction motor cooling inverter)		2F_05.01.Zefiro-Europe.CONCEPT.657

Bus Interface

TCMS SW to Propulsion supplier:

Nr	Functional name	Type	Linked to requirement
2F_05.01.Zefiro-Europe.CONCEPT.403	Car identity	A	2F_05.01.Zefiro-Europe.CONCEPT.105
2F_05.01.Zefiro-Europe.CONCEPT.404	LCM configuration data: – Position of active LCM in unit – Number of active LCMs in unit – Unit position in train – Total number of units in train	A	2F_05.01.Zefiro-Europe.CONCEPT.239
2F_05.01.Zefiro-Europe.CONCEPT.405	Neutral section active	L	2F_05.01.Zefiro-Europe.CONCEPT.210
	Separation section active		2F_05.01.Zefiro-Europe.CONCEPT.210
2F_05.01.Zefiro-Europe.CONCEPT.407	Enable start of motor converter	L	2F_05.01.Zefiro-Europe.CONCEPT.204
2F_05.01.Zefiro-Europe.CONCEPT.408	Enable towing mode	L	2F_05.01.Zefiro-Europe.CONCEPT.210
2F_05.01.Zefiro-Europe.CONCEPT.409	Cut-out LCM	L	2F_05.01.Zefiro-Europe.CONCEPT.145
2F_05.01.Zefiro-Europe.CONCEPT.448	Cut-out ACM	L	2F_05.01.Zefiro-Europe.CONCEPT.145
2F_05.01.Zefiro-Europe.CONCEPT.449	Cut-out MCM1	L	2F_05.01.Zefiro-Europe.CONCEPT.145
2F_05.01.Zefiro-Europe.CONCEPT.450	Cut-out MCM2	L	2F_05.01.Zefiro-Europe.CONCEPT.145
2F_05.01.Zefiro-Europe.CONCEPT.451	Cut-out FCM	L	2F_05.01.Zefiro-Europe.CONCEPT.145
2F_05.01.Zefiro-Europe.CONCEPT.421	Train weight correction factor	A	2F_05.01.Zefiro-Europe.CONCEPT.184
2F_05.01.Zefiro-Europe.CONCEPT.422	Emergency brake is active	L	2F_05.01.Zefiro-Europe.CONCEPT.430
2F_05.01.Zefiro-Europe.CONCEPT.630	Start Automatic test	L	2F_05.01.Zefiro-Europe.CONCEPT.150
	Line trip test		2F_05.01.Zefiro-Europe.CONCEPT.678
2F_05.01.Zefiro-Europe.CONCEPT.634	Transient recorder activation trigg	L	2F_05.01.Zefiro-Europe.CONCEPT.140
2F_05.01.Zefiro-Europe.CONCEPT.677	Traction motor cooling inverter ok		2F_05.01.Zefiro-Europe.CONCEPT.655 2F_05.01.Zefiro-Europe.CONCEPT.671
2F_05.01.Zefiro-Europe.CONCEPT.HR50	Rate Reference		2F_05.01.Zefiro-Europe.CONCEPT.HR04
2F_05.01.Zefiro-Europe.CONCEPT.HR51	Vehicle speed		2F_05.01.Zefiro-Europe.CONCEPT.HR04
2F_05.01.Zefiro-Europe.CONCEPT.HR52	Medium voltage Available		2F_05.01.Zefiro-Europe.CONCEPT.HR04

Propulsion supplier to TCMS SW:

Nr	Functional name	Type	Linked to requirement
2F_05.01.Zefiro-Europe.CONCEPT.33	Converter status: LCM is in operation ACM is in operation FCM is in operation MCM1 is in operation MCM2 is in operation LCM is cut-out ACM is cut-out FCM is cut-out MCM1 is cut-out MCM2 is cut-out LCM is faulty ACM is faulty FCM is faulty MCM1 is faulty MCM2 is faulty	L	2F_05.01.Zefiro-Europe.CONCEPT.252
2F_05.01.Zefiro-Europe.CONCEPT.626	Mechanical fault MCM1	L	2F_05.01.Zefiro-Europe.CONCEPT.615
2F_05.01.Zefiro-Europe.CONCEPT.627	Mechanical fault MCM2	L	2F_05.01.Zefiro-Europe.CONCEPT.615
2F_05.01.Zefiro-Europe.CONCEPT.HR53	Propulsion back direction		2F_05.01.Zefiro-Europe.CONCEPT.HR01
2F_05.01.Zefiro-Europe.CONCEPT.HR54	WSP status: WSP_Hold, WSP_Vent		2F_05.01.Zefiro-Europe.CONCEPT.268
2F_05.01.Zefiro-Europe.CONCEPT.629	Automatic test in progress	L	2F_05.01.Zefiro-Europe.CONCEPT.628
2F_05.01.Zefiro-Europe.CONCEPT.660	Transformer is isolated		2F_05.01.Zefiro-Europe.CONCEPT.659
2F_05.01.Zefiro-Europe.CONCEPT.676	Activate cooling for Traction motor		2F_05.01.Zefiro-Europe.CONCEPT.657 2F_05.01.Zefiro-Europe.CONCEPT.670

Brake to TCMS SW:

Nr	Functional name	Type	Linked to requirement
2F_05.01.Zefiro-Europe.CONCEPT.70	Load pressure bogie 1		
2F_05.01.Zefiro-Europe.CONCEPT.70	Load pressure bogie 2		
2F_05.01.Zefiro-Europe.CONCEPT.633	Load transducer faulty bogie 1		
2F_05.01.Zefiro-Europe.CONCEPT.633	Load transducer faulty bogie 2		